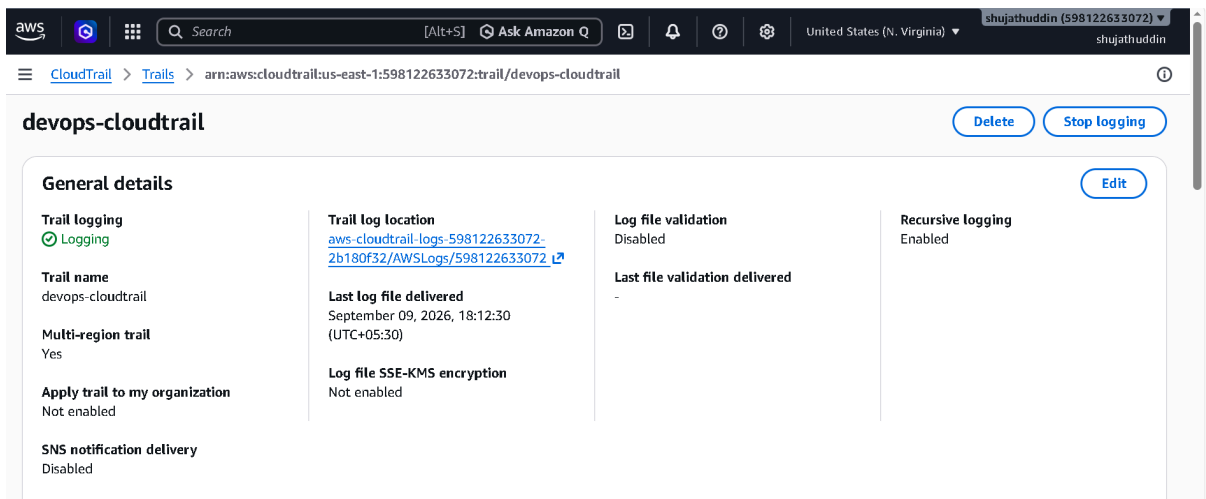


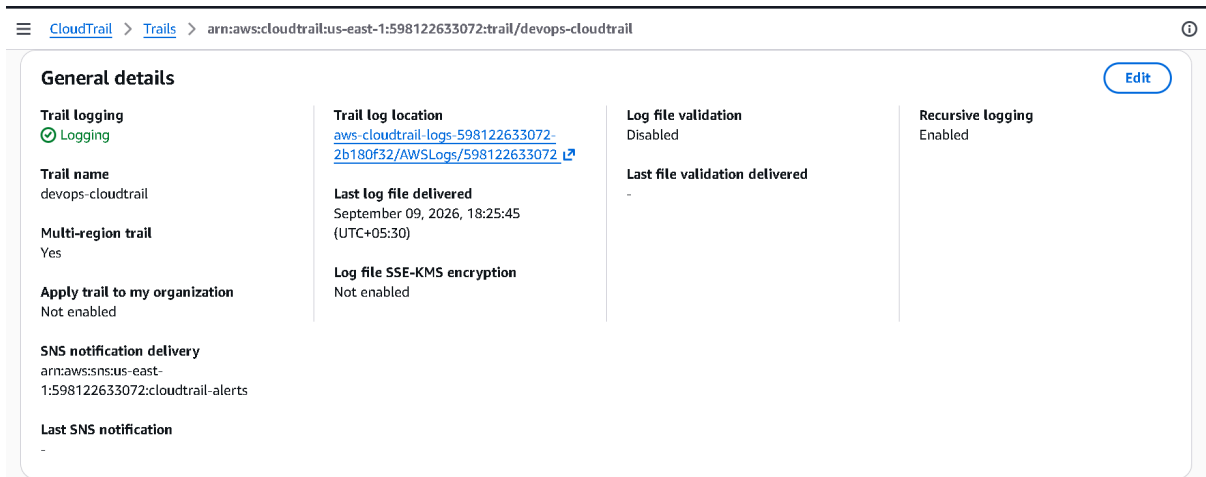
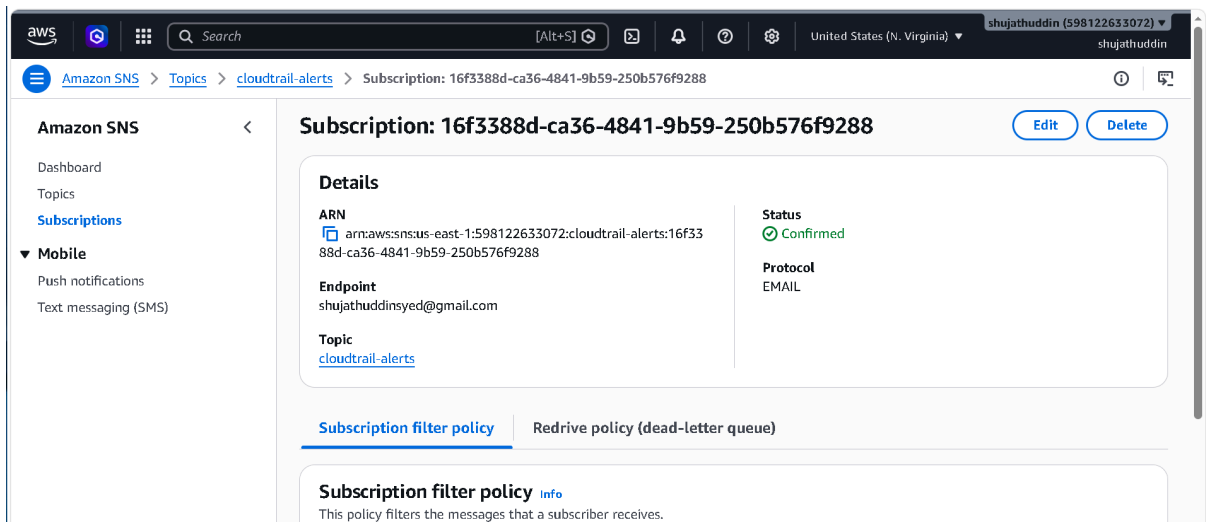
1. Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events.

1. Opened **AWS Console** → **CloudTrail**.
2. Created a new CloudTrail trail.
3. Provided a trail name.
4. Created an **S3 bucket** to store CloudTrail log files.
5. Enabled **Management Events** for Read and Write operations.
6. Enabled **CloudWatch Logs** integration.
7. Created a CloudWatch Log Group for CloudTrail events.
8. Configured the required IAM role.
9. Created the CloudTrail trail.
10. Verified that CloudTrail events were being stored in **S3** and **CloudWatch Logs**.



2. Enable SNS for CloudTrail to send alerts via email.

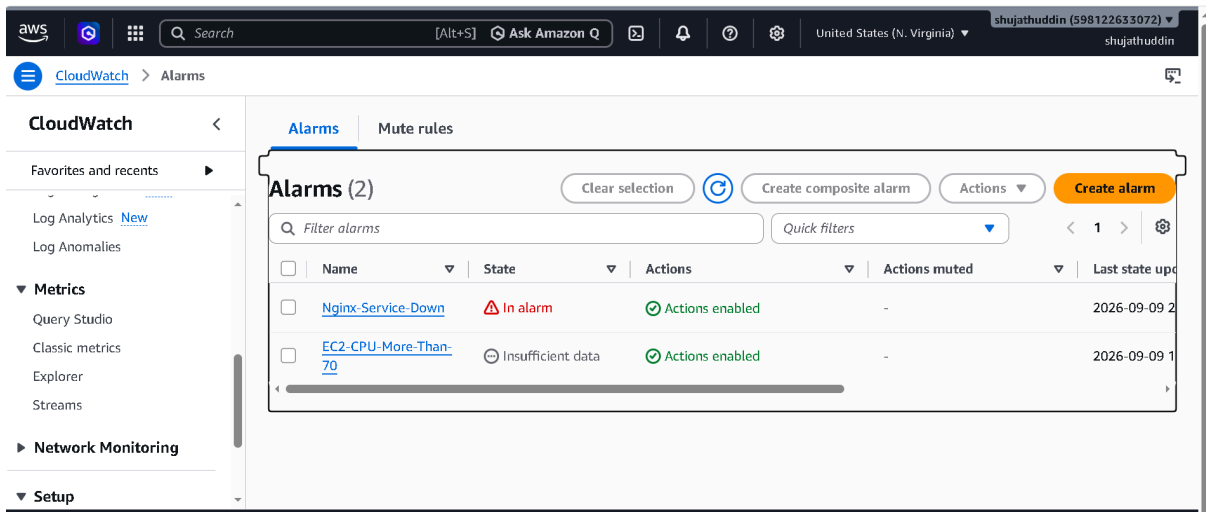
- A. Opened **AWS Console** → **Amazon SNS**.
- B. Created an SNS topic named **cloudtrail-alerts**.
- C. Created an **Email subscription** for the required email address.
- D. Confirmed the subscription using the confirmation email.
- E. Connected the SNS topic with CloudTrail notifications.
- F. Verified that CloudTrail notifications could be delivered through SNS.
- G. Tested the configuration and verified email notification delivery.



3. Configure CloudWatch monitoring and record the CPU utilization and other metrics of EC2.

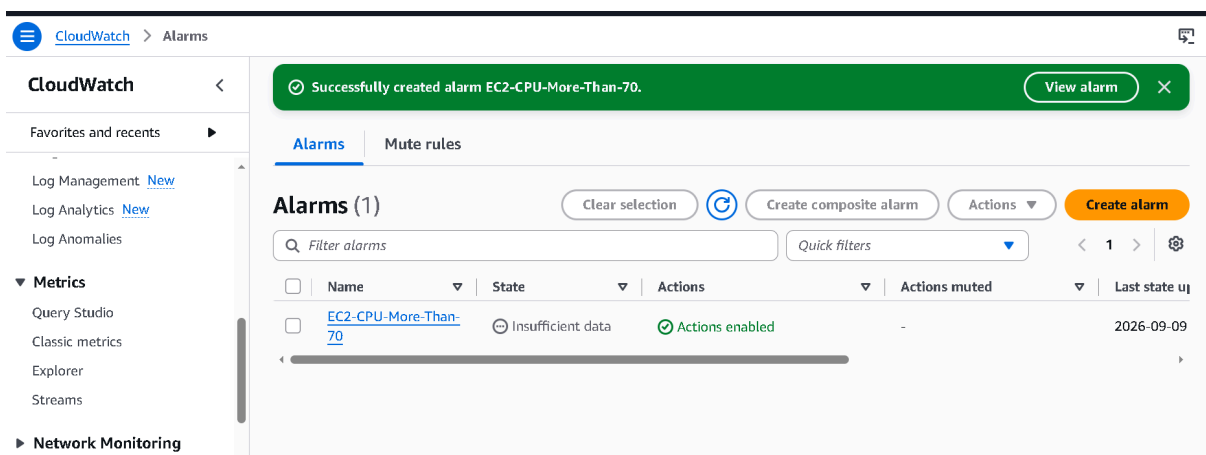
1. Opened **AWS Console** → **EC2**.
2. Selected the required EC2 instance.
3. Attached an IAM role with **CloudWatchAgentServerPolicy** permission.
4. Connected to the EC2 instance using SSH.
5. Installed the **Amazon CloudWatch Agent**.
6. Configured the CloudWatch Agent to collect system metrics.
7. Started the CloudWatch Agent.
8. Opened **CloudWatch** → **Metrics**.
9. Verified EC2 metrics such as:
10. CPU utilization
11. Memory utilization
12. Disk utilization
13. Network traffic
14. Status checks

15. Confirmed that the metrics were being recorded in CloudWatch.



4. Create one alarm to send an alert to email if the CPU utilization is more than 70 percent.

- Opened **CloudWatch** → **Alarms**.
- Selected **Create alarm**.
- Selected the EC2 instance's **CPUUtilization** metric.
- Set the statistic to **Average**.
- Set the threshold condition to **CPU utilization > 70%**.
- Configured an SNS topic for email notifications.
- Named the alarm **EC2-CPU-More-Than-70**.
- Created the alarm.
- Tested the alarm by increasing CPU utilization.
- Verified that an email alert was received when CPU utilization exceeded 70%.



5. Create a Dashboard and monitor the Tomcat service whether it is running or not and send the alert.

1. Opened **CloudWatch** → **Dashboards**.
2. Created a dashboard named **DevOps-Services-Dashboard**.
3. Added a metric widget for the Tomcat/Java process.
4. Configured CloudWatch Agent **procstat** monitoring for the Tomcat process.
5. Added the Tomcat process-count metric to the dashboard.
6. Created a CloudWatch alarm for the Tomcat process.
7. Set the alarm condition to trigger when the Tomcat process count becomes **less than 1**.
8. Configured SNS email notification.
9. Named the alarm **Tomcat-Service-Down**.
10. Tested the alarm by stopping Tomcat.
11. Verified that the dashboard showed the Tomcat process as unavailable and an email alert was received.
12. Started Tomcat again and verified that the service returned to normal.

← ↻ Not secure 18.235.2.1:8080 ☆ ☆ ⚙ Chat

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Apache Tomcat/11.0.11

 **APACHE**™ SOFTWARE FOUNDATION
<http://www.apache.org/>

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:
[Security Considerations How-To](#)
[Manager Application How-To](#)
[Clustering/Session Replication How-To](#)

[Server Status](#)
[Manager App](#)
[Host Manager](#)

Developer Quick Start

[Tomcat Setup](#)
[First Web Application](#)

[Realms & AAA](#)
[JDBC DataSources](#)

[Examples](#)

[Servlet Specifications](#)
[Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 11.0 access to the manager

Documentation

[Tomcat 11.0 Documentation](#)
[Tomcat 11.0 Configuration](#)
[Tomcat Wiki](#)

Getting Help

[FAQ and Mailing Lists](#)

The following mailing lists are available:

[tomcat-announce](#)
Important announcements, releases, security

```

ec2-user@ip-20-0-1-164 ~]$ aws cloudwatch list-metrics --namespace TomcatMonitoring

  "Metrics": [
    {
      "Namespace": "TomcatMonitoring",
      "MetricName": "TomcatStatus",
      "Dimensions": []
    }
  ]

ec2-user@ip-20-0-1-164 ~]$ aws cloudwatch get-metric-statistics \
--namespace TomcatMonitoring \
--metric-name TomcatStatus \
--statistics Maximum \
--period 300 \
--start-time $(date -u -d '10 minutes ago' +%Y-%m-%dT%H:%M:%SZ) \
--end-time $(date -u +%Y-%m-%dT%H:%M:%SZ)

  "Label": "TomcatStatus",
  "Datapoints": [
    {
      "Timestamp": "2026-09-09T14:39:00+00:00",
      "Maximum": 1.0,
      "Unit": "Count"
    }
  ]

ec2-user@ip-20-0-1-164 ~]$

```

6. Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.

1. Opened **CloudWatch** → **Dashboards**.
2. Added a metric widget for the Nginx process.
3. Configured CloudWatch Agent **procstat** monitoring for Nginx.
4. Added the Nginx process-count metric to the dashboard.
5. Created a CloudWatch alarm for the Nginx process.
6. Set the alarm condition to trigger when the Nginx process count becomes **less than 1**.
7. Configured SNS email notification.
8. Named the alarm **Nginx-Service-Down**.
9. Tested the alarm by stopping Nginx.
10. Verified that the dashboard showed Nginx as unavailable and an email alert was received.
11. Started Nginx again and verified that the service returned to normal

```

Last login: Wed Sep  9 14:45:39 2026 from 18.206.107.29
[ec2-user@ip-20-0-1-164 ~]$ sudo crontab -l
* * * * * /usr/local/bin/check-nginx.sh
[ec2-user@ip-20-0-1-164 ~]$ aws cloudwatch list-metrics --namespace NginxMonitoring
{
  "Metrics": [
    {
      "Namespace": "NginxMonitoring",
      "MetricName": "NginxStatus",
      "Dimensions": []
    }
  ]
}
[ec2-user@ip-20-0-1-164 ~]$
```

